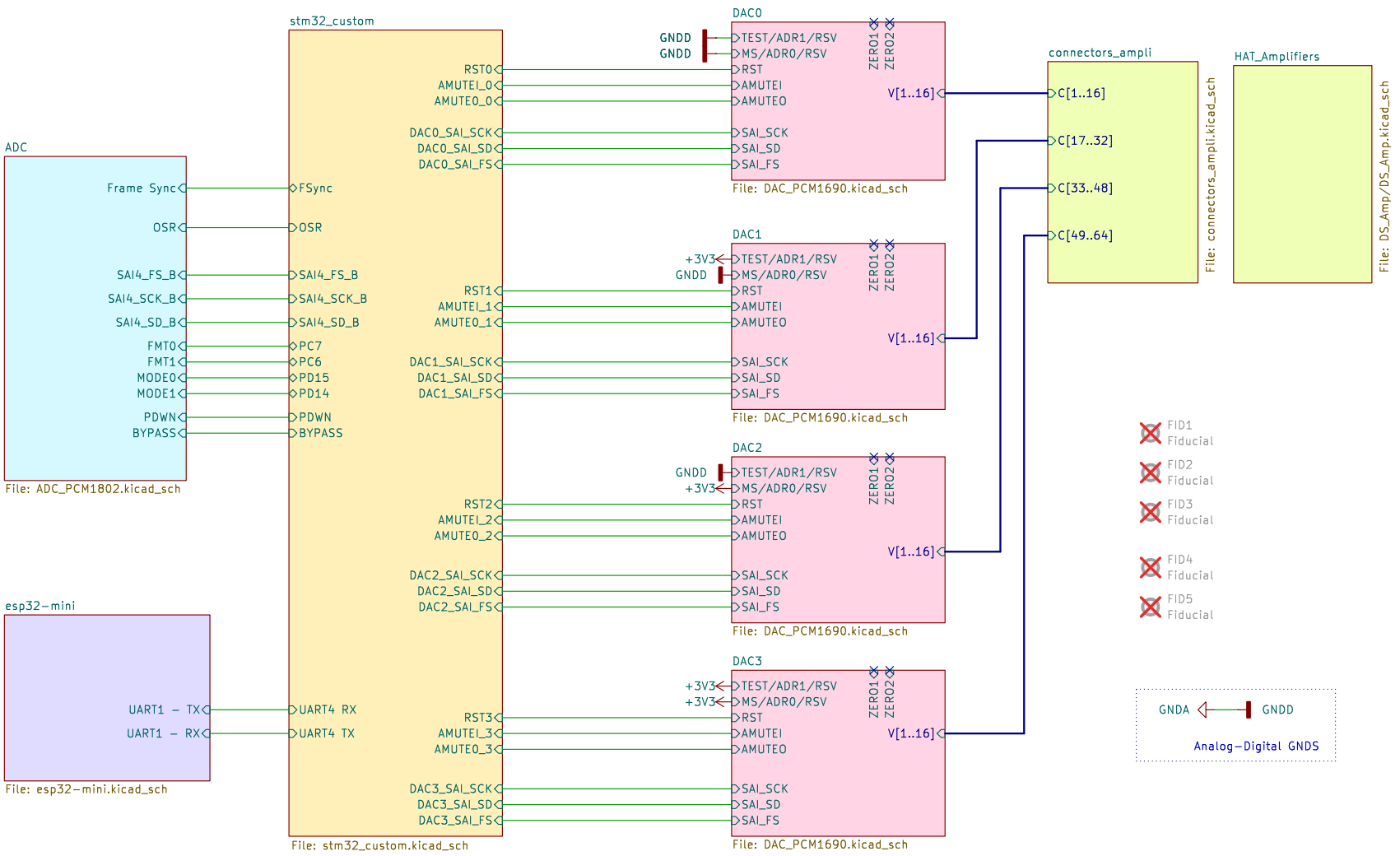
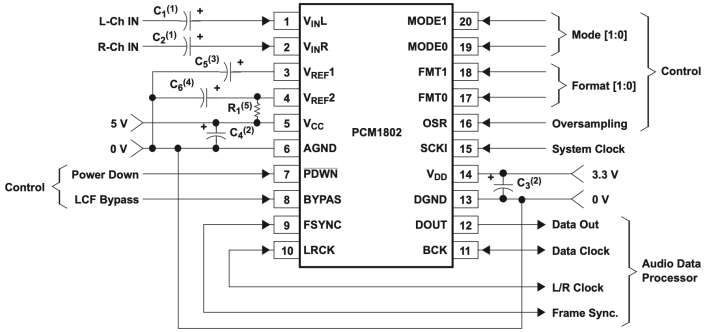


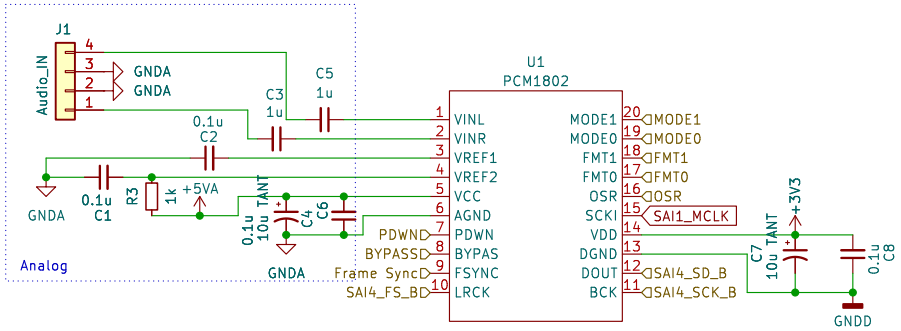
Sheet: /
File: DS.kicad_sch

Title: Master

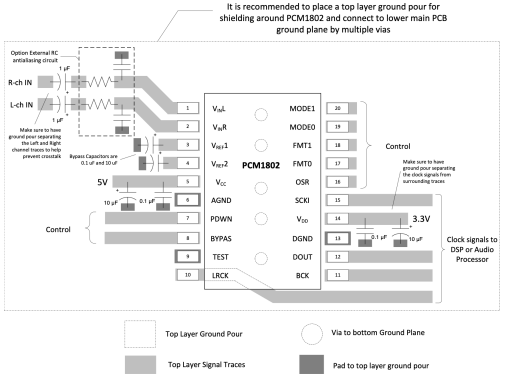
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 1/29



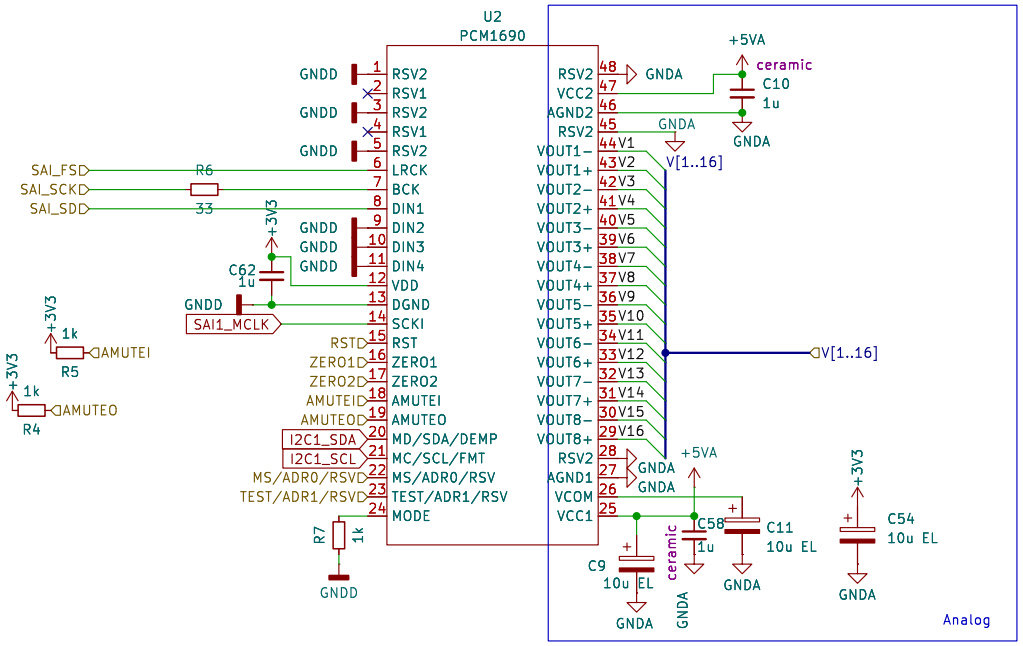
Copyright © 2016, Texas Instruments Incorporated

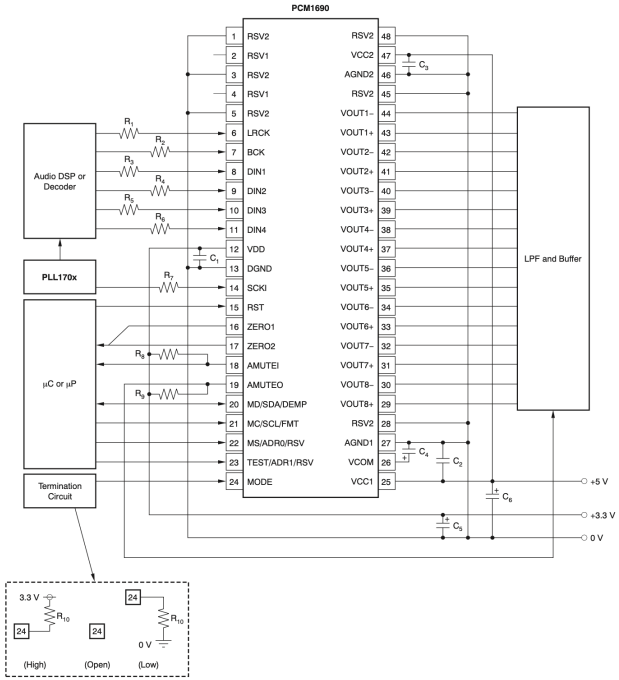


10.2 Layout Example



Sheet: /ADC/		
File: ADC_PCM1802.kicad_sch		
Title: Conversor ADC PCM1802		
Size: A4	Date: 2026-02-01	**Rev:**
KiCad E.D.A. 9.0.7		Id: 2/29

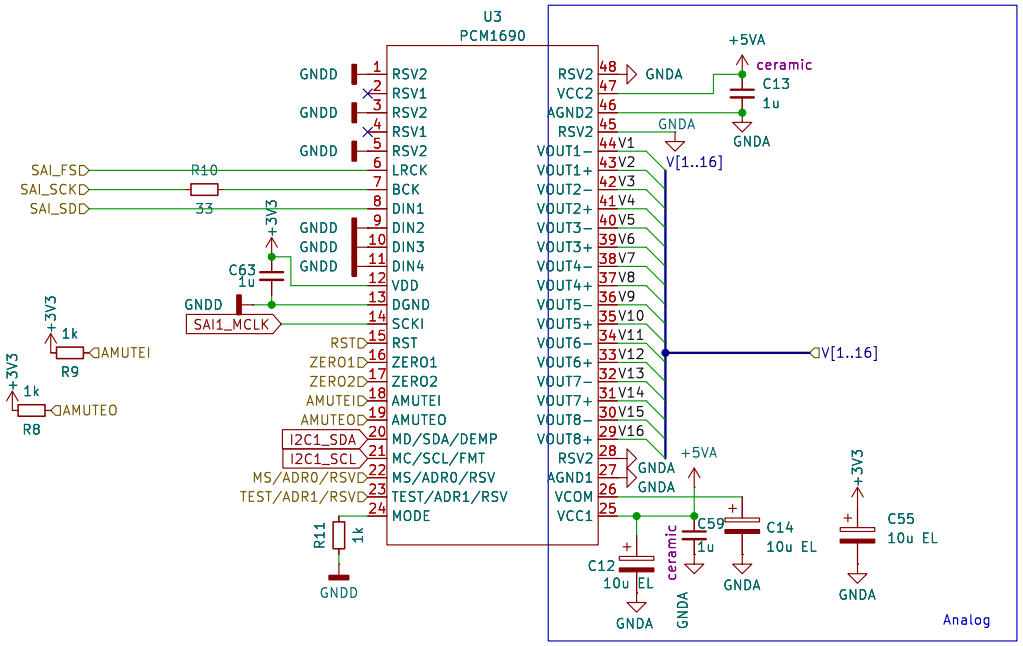


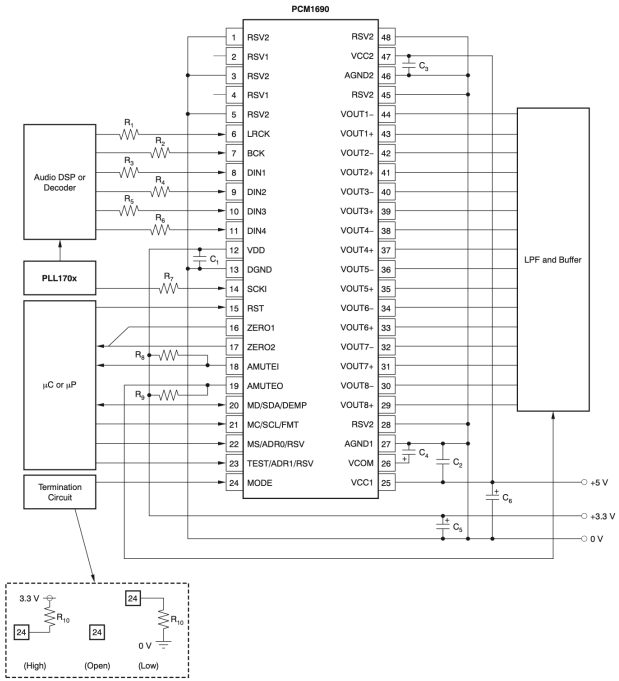


NOTE: C_1 through C_3 are 1-µF ceramic capacitors. C_4 through C_6 are 10-µF electrolytic capacitors. R_1 through R_7 are 22-Ω to 100-Ω resistors. R_8 and R_9 are resistors appropriate for pull-up. R_{10} is less than 10 kΩ.

Figure 41. Basic Connection Diagram

Sheet: /DAC2/		
File: DAC_PCM1690.kicad_sch		
Title: Conversores DAC PCM1690 8−channel		
Size: A4	Date: 2026−02−01	**Rev:**
KiCad E.D.A. 9.0.7		Id: 3/29

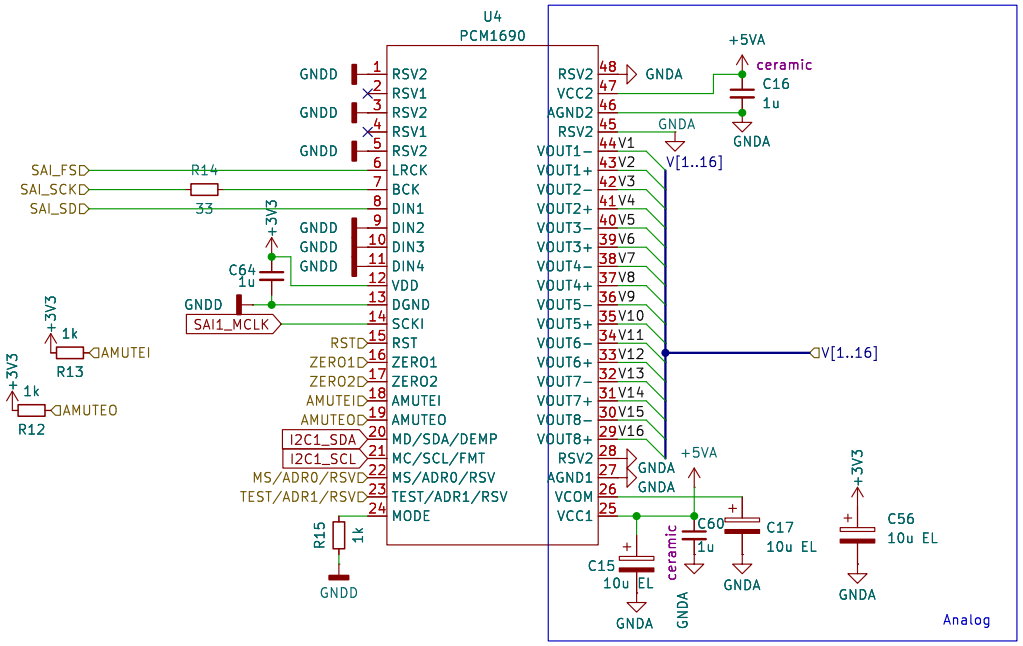


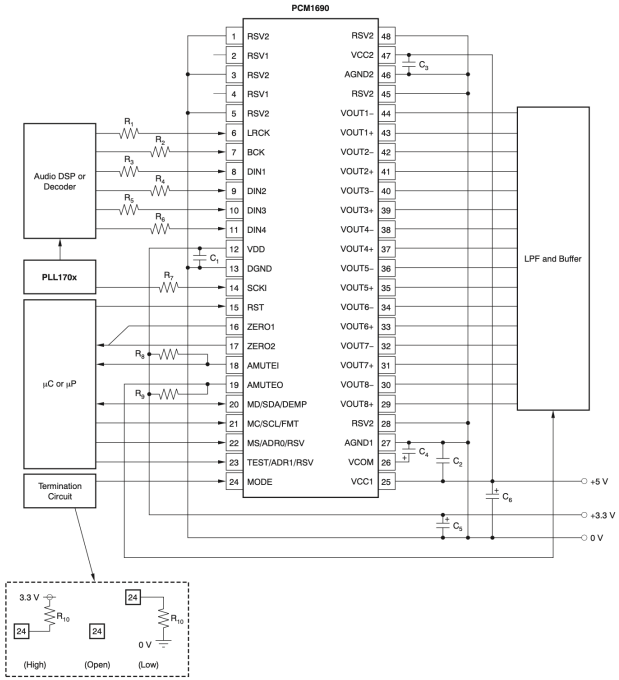


NOTE: C_1 through C_3 are 1-µF ceramic capacitors. C_4 through C_6 are 10-µF electrolytic capacitors. R_1 through R_7 are 22-Ω to 100-Ω resistors. R_8 and R_9 are resistors appropriate for pull-up. R_{10} is less than 10 kΩ.

Figure 41. Basic Connection Diagram

Sheet: /DAC0/		
File: DAC_PCM1690.kicad_sch		
Title: Conversores DAC PCM1690 8−channel		
Size: A4	Date: 2026−02−01	**Rev:**
KiCad E.D.A. 9.0.7		Id: 4/29

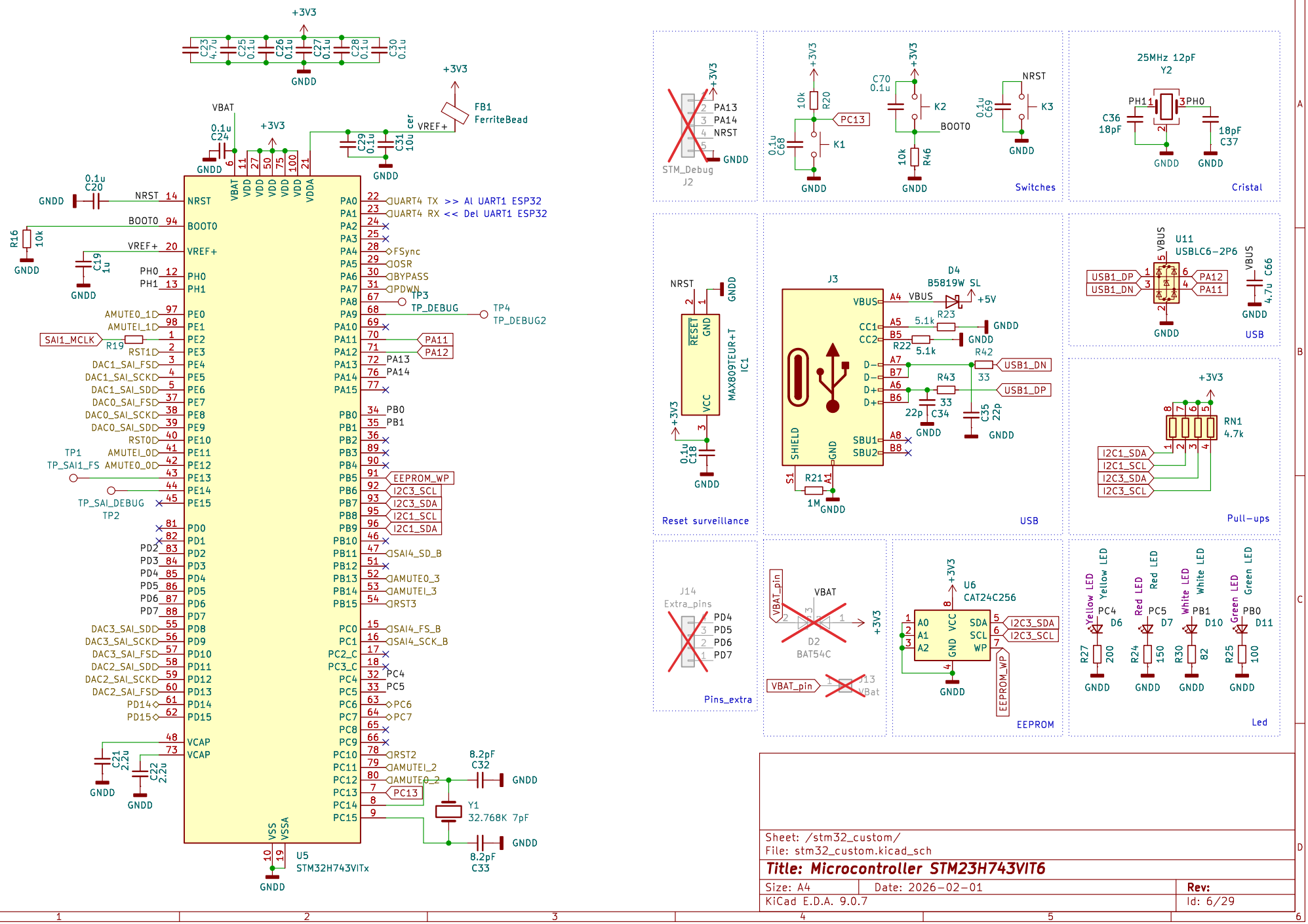


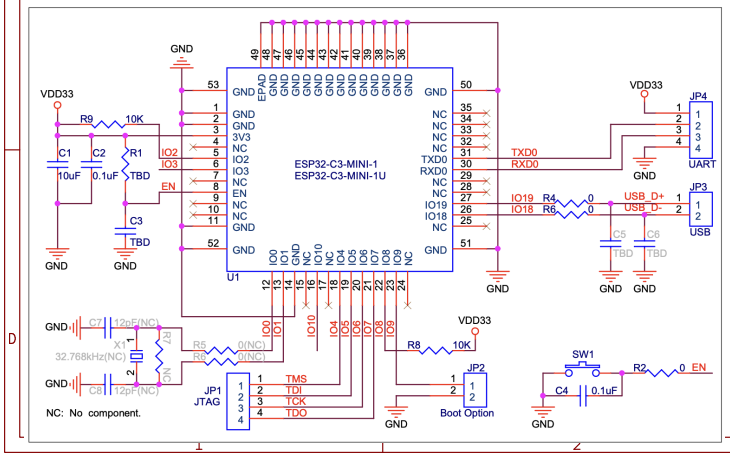


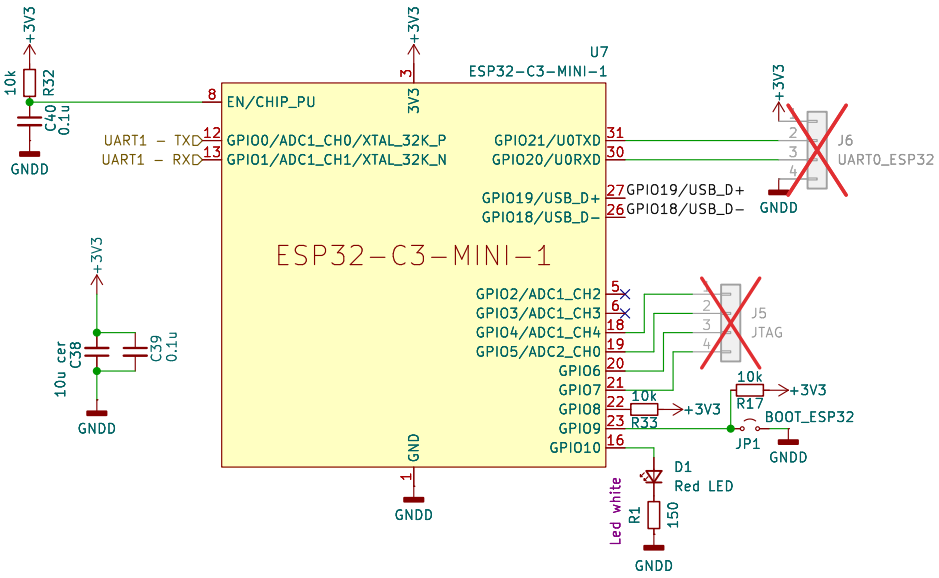
NOTE: C_1 through C_3 are 1-µF ceramic capacitors. C_4 through C_6 are 10-µF electrolytic capacitors. R_1 through R_7 are 22-Ω to 100-Ω resistors. R_8 and R_9 are resistors appropriate for pull-up. R_{10} is less than 10 kΩ.

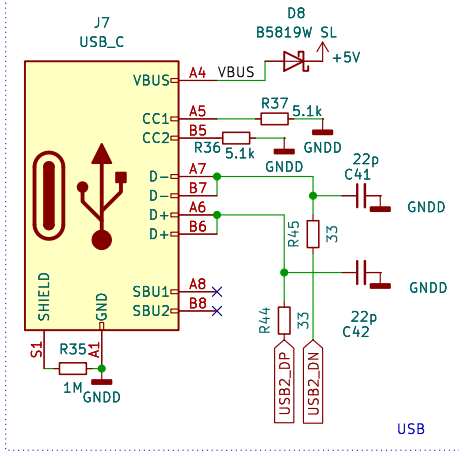
Figure 41. Basic Connection Diagram

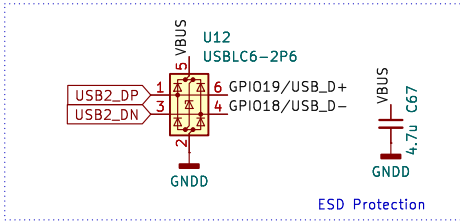
Sheet: /DAC1/		
File: DAC_PCM1690.kicad_sch		
Title: Conversores DAC PCM1690 8−channel		
Size: A4	Date: 2026−02−01	**Rev:**
KiCad E.D.A. 9.0.7		Id: 5/29

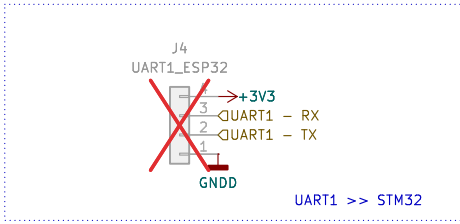

+3V3
GNDD
VBAT
FB1
FerriteBead
VREF+
U5
STM32H743VITx
NRST
BOOT0
PH0
PH1
SAI1_MCLK
TP_DEBUG
TP_SAI_DEBUG
TP_SAI1_FS
PA11
PA12
EEPROM_WP
I2C3_SCL
I2C3_SDA
I2C1_SCL
I2C1_SDA
PC13
PC4
PC5
PC6
PC7
PA13
PA14
PB0
PB1
UART4 TX >> Al UART1 ESP32
UART4 RX << Del UART1 ESP32
FSync
OSR
BYPASS
PDWN
SAI4_SD_B
AMUTE0_3
AMUTEI_3
RST3
SAI4_FS_B
SAI4_SCK_B
RST2
AMUTEI_2
AMUTE0_2
STM_Debug
J2
Switches
Cristal
25MHz 12pF
Y2
Reset surveillance
MAX809TEUR+T
IC1
USB
J3
B5819W SL
USBLC6-2P6
U11
Pull-ups
RN1
4.7k
Pins_extra
J14
Extra_pins
BAT54C
EEPROM
U6
CAT24C256
Led
Yellow LED
Red LED
White LED
Green LED
Sheet: /stm32_custom/
File: stm32_custom.kicad_sch
Title: Microcontroller STM23H743VIT6
Size: A4
Date: 2026-02-01
KiCad E.D.A. 9.0.7
Rev:
Id: 6/29







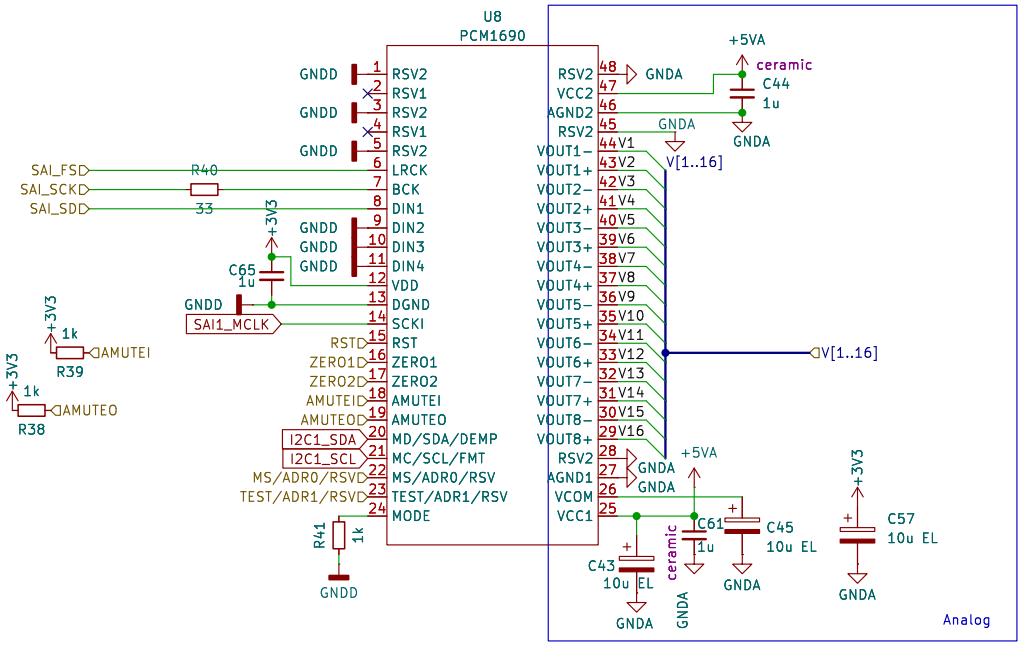


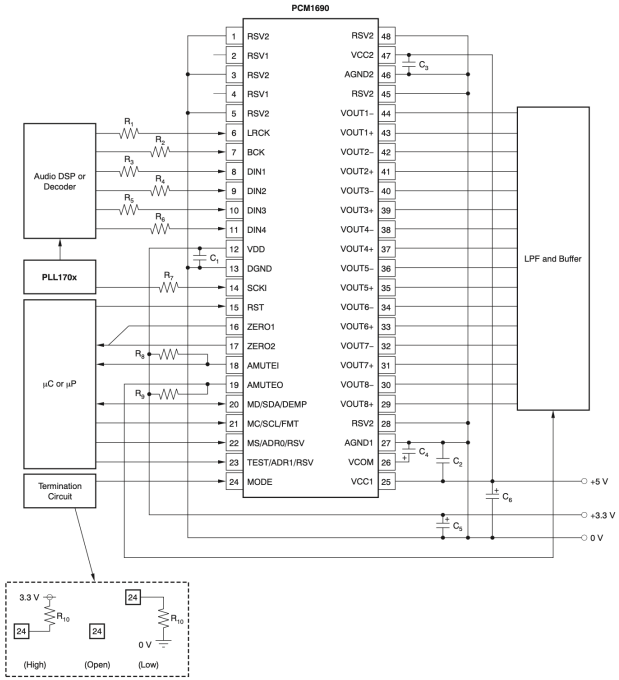


Sheet: /esp32-mini/
File: esp32-mini.kicad_sch

Title: ESP32

Size: A4	Date: 2026-02-01	**Rev:**
KiCad E.D.A. 9.0.7		Id: 7/29

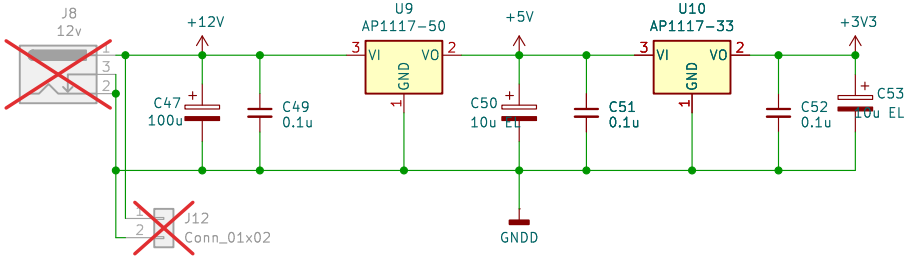




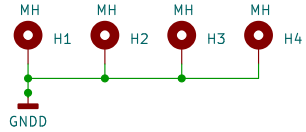
NOTE: C_1 through C_3 are 1-µF ceramic capacitors. C_4 through C_6 are 10-µF electrolytic capacitors. R_1 through R_7 are 22-Ω to 100-Ω resistors. R_8 and R_9 are resistors appropriate for pull-up. R_{10} is less than 10 kΩ.

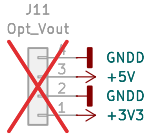
Figure 41. Basic Connection Diagram

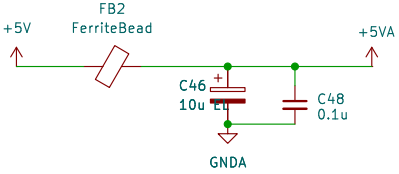
Sheet: /DAC3/		
File: DAC_PCM1690.kicad_sch		
Title: Conversores DAC PCM1690 8−channel		
Size: A4	Date: 2026−02−01	**Rev:**
KiCad E.D.A. 9.0.7		Id: 8/29







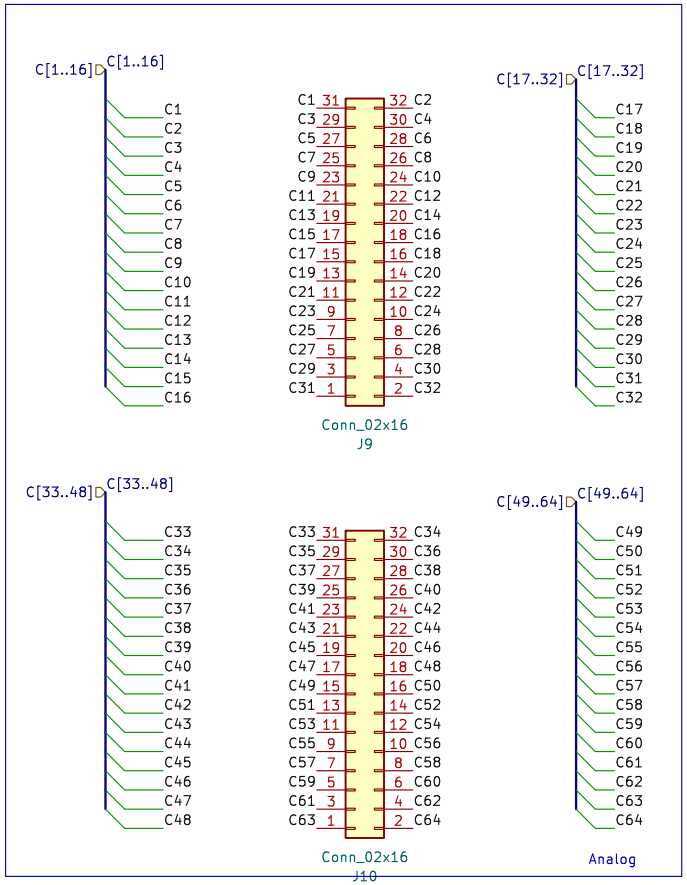




Sheet: /DS_Power/
File: DS_Power.kicad_sch

Title: Power supply 12V

Size: A4	Date: 2026-02-01	**Rev:**
KiCad E.D.A. 9.0.7		Id: 9/29



Sheet: /connectors_ampli/
File: connectors_ampli.kicad_sch

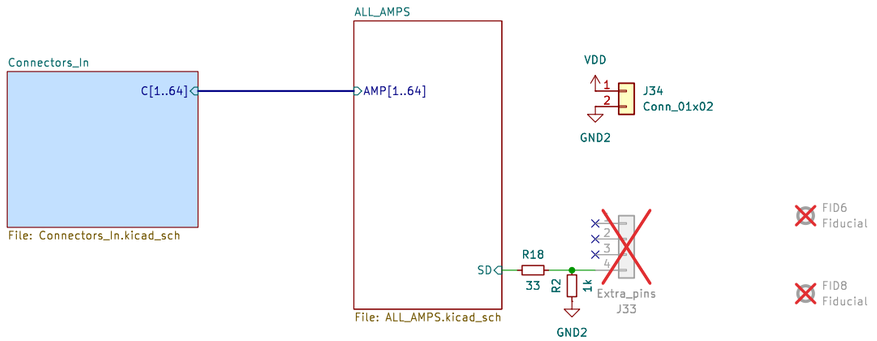
Title: Connectors

Size: A4	Date: 2026-02-01	**Rev:**
KiCad E.D.A. 9.0.7		Id: 9/29

DS_Amp – Hat Amplificador 32 Canales

Arquitectura:
– 16x PAM8403 (dual Class-D filterless)
– Entrada diferencial desde PCM1690 via J9/J10
– 32 parlantes 0.5–1W / 8 ohm
– PSU: 5V / 10A externa (separada de DS9)

Bancos:
– Bank A: Ch 1–8   (DAC0, J9 pins 1–16)
– Bank B: Ch 9–16 (DAC1, J9 pins 17–32)
– Bank C: Ch 17–24 (DAC2, J10 pins 1–16)
– Bank D: Ch 25–32 (DAC3, J10 pins 17–32)



5V / 10A PSU externa
Para DS9 Beamforming System
32x PAM8403 Class-D Filterless

Sheet: /HAT_Amplifiers/
File: DS_Amp.kicad_sch

Title: DS_Amp – 32ch Amplifier Hat

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 11/29

INPUT CONNECTORS

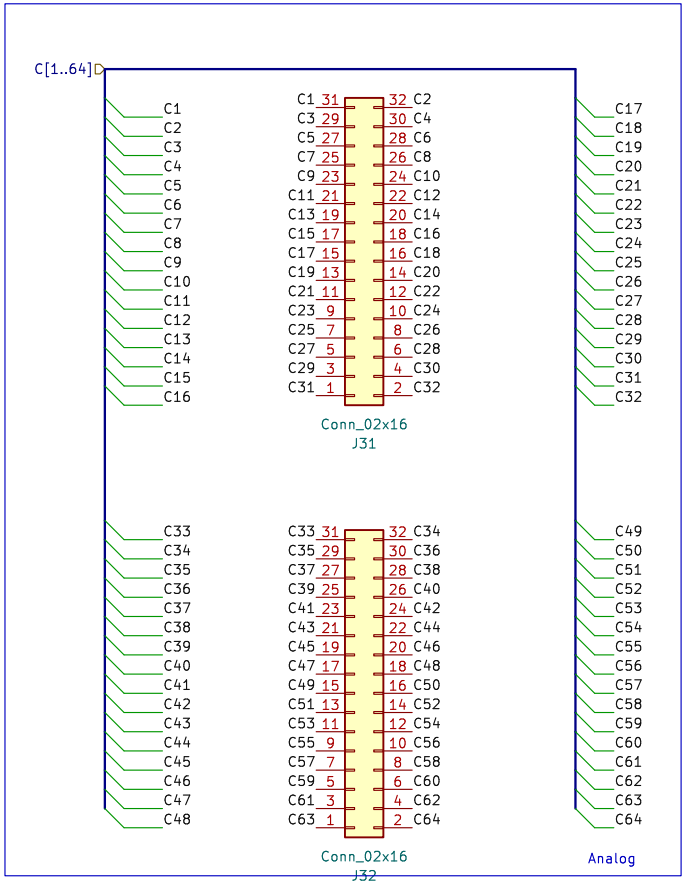
J1: PinHeader 2x16 macho (mates J9 en DS9)
    32 pines = 16 pares diferenciales
    Ch 1–8 (DAC0) + Ch 9–16 (DAC1)

J2: PinHeader 2x16 macho (mates J10 en DS9)
    32 pines = 16 pares diferenciales
    Ch 17–24 (DAC2) + Ch 25–32 (DAC3)

J_GND: Terminal block o pad
    Conexion GND robusta entre DS9 y hat
    CRITICO: Debe ser conexion gruesa

Footprint PinHeader: Connector_PinHeader_2.54mm:PinHeader_2x16_P2.54mm_Vertical

Pinout segun DS9 connectors_ampli.kicad_sch:
    Pines impares (fila 1): VOUT+ canales
    Pines pares (fila 2): VOUT− canales

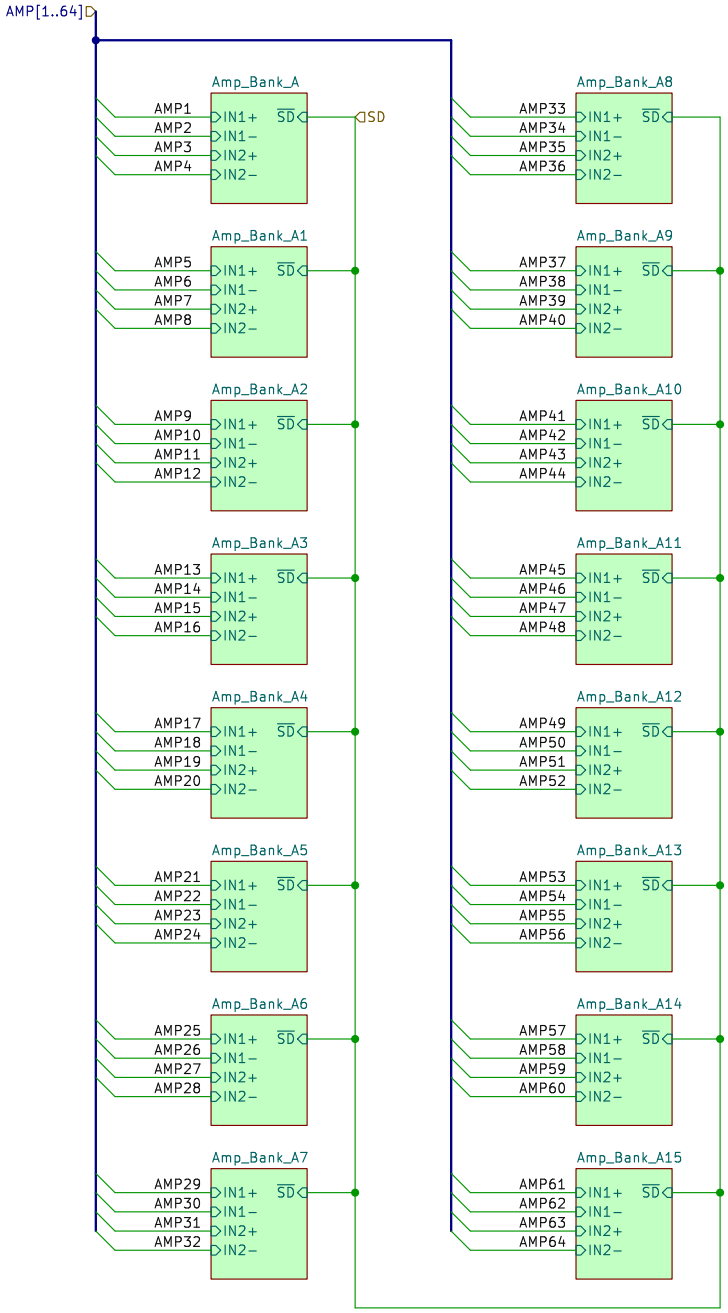


64 senales audio diferencial + GND bus
2x PinHeader 2x16 macho (mates DS9 J9/J10)

Sheet: /HAT_Amplifiers/Connectors_In/
File: Connectors_In.kicad_sch

Title: DS_Amp — Input Connectors

Size: A4	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 11/29



Sheet: /HAT_Amplifiers/ALL_AMPS/
File: ALL_AMPS.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 16/29

AMPLIFIER BANK — 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

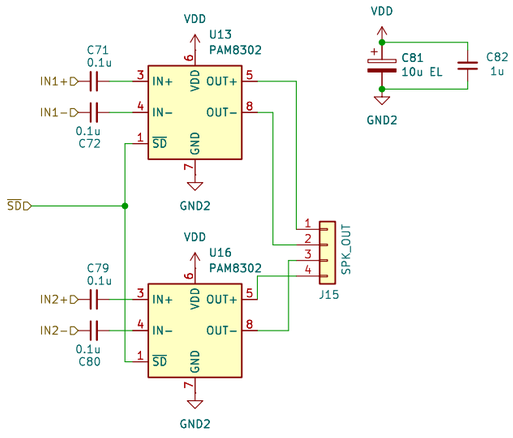
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A1/
File: Amp_Bank.kicad_sch

Title: DS_Amp — Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 12/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

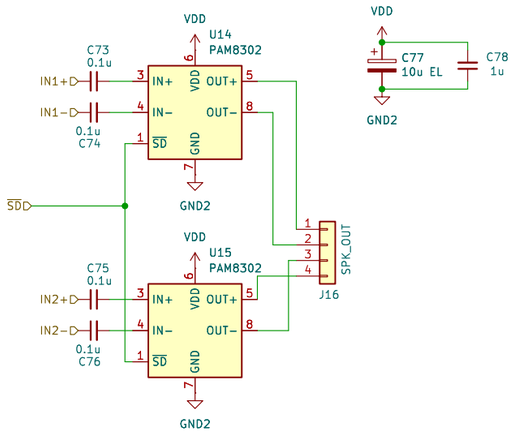
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 14/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

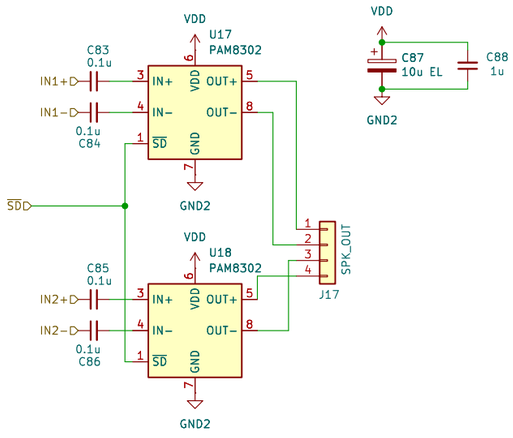
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A2/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 15/29

AMPLIFIER BANK — 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

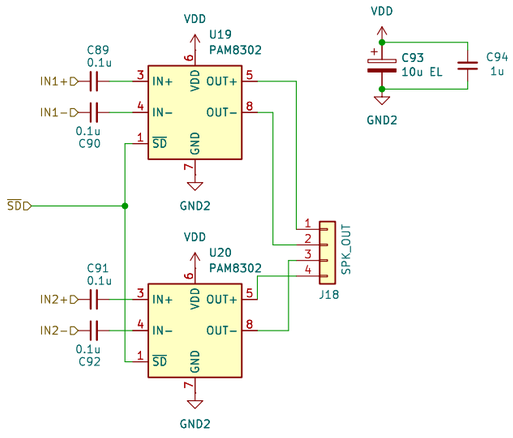
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A3/
File: Amp_Bank.kicad_sch

Title: DS_Amp — Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 17/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

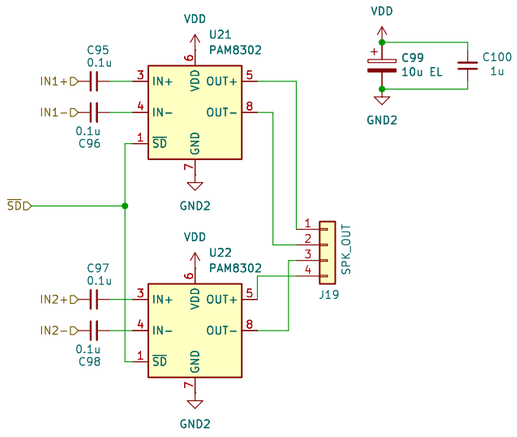
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A4/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 18/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

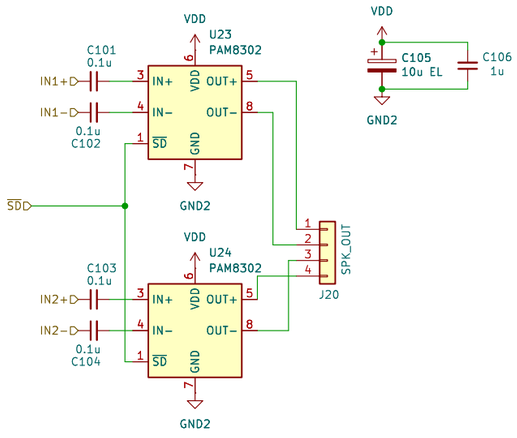
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A5/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 19/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

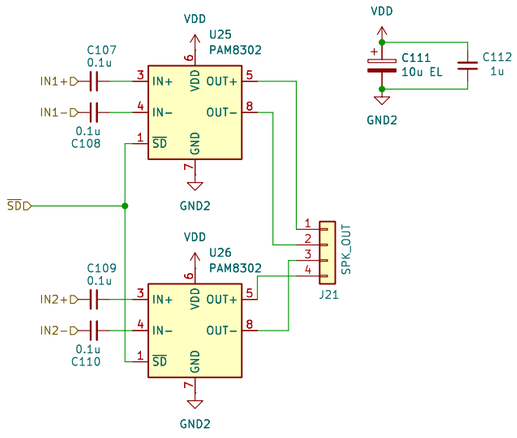
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A6/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 20/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

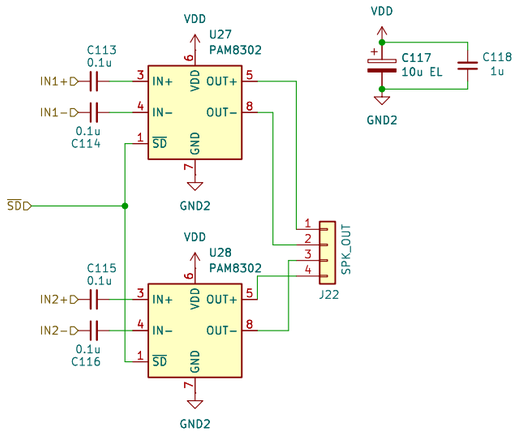
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A7/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 21/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

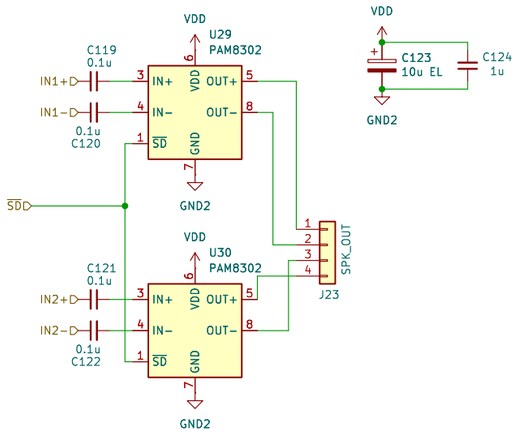
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A8/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 22/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

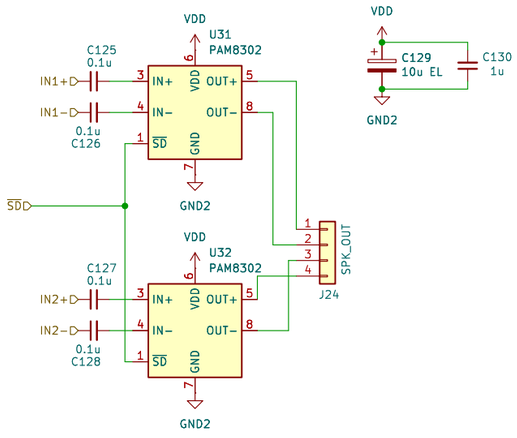
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A9/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 23/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

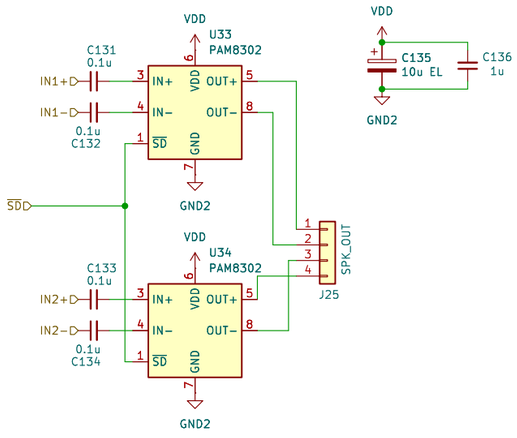
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A10/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 24/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

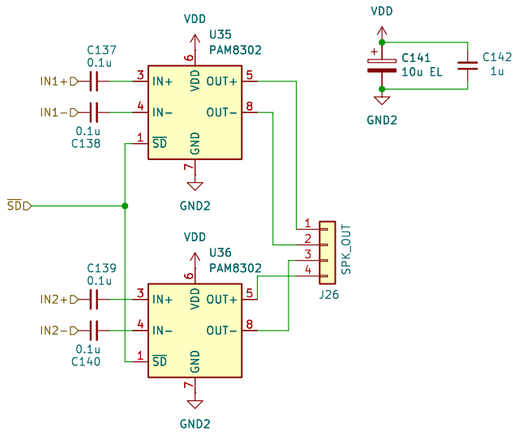
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A11/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 25/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

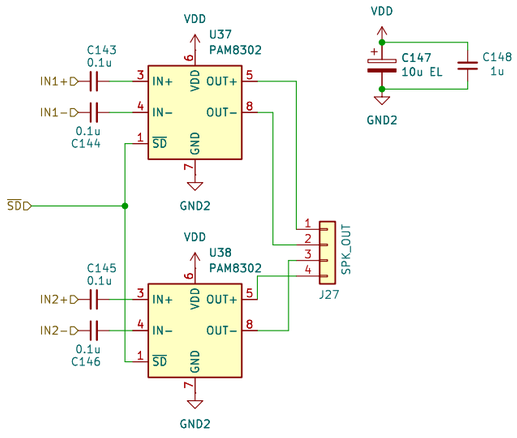
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A12/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 26/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

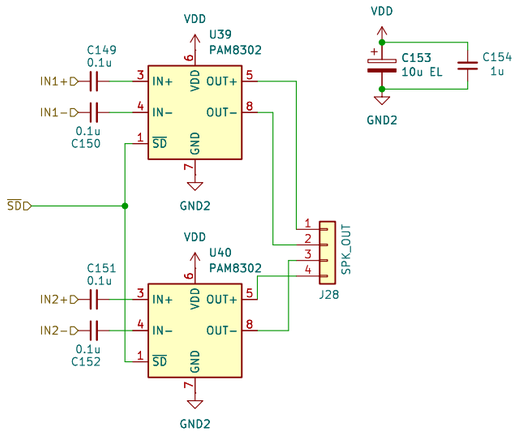
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A13/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 27/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

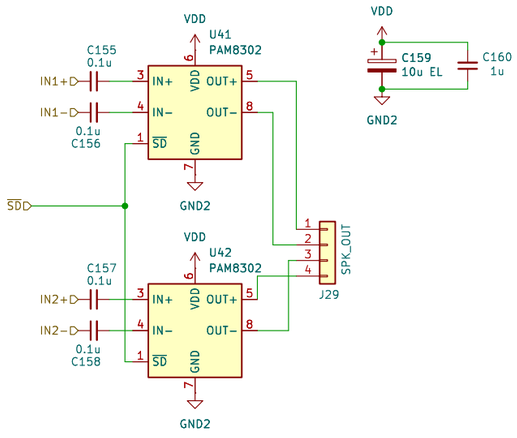
Total por banco: 8 ICs + 32 pasivos

PAM8302A Pinout (MSOP-8):

```
 Pin 1: SD   (shutdown, HIGH=on)
 Pin 2: NC   (no connect)
 Pin 3: IN+  (entrada diferencial +)
 Pin 4: IN-  (entrada diferencial -)
 Pin 5: VO+  (salida BTL +)
 Pin 6: VDD  (2.5V - 5.5V)
 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
Salida: BTL (no conectar parlante a GND)

LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A14/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 28/29

AMPLIFIER BANK – 8 CANALES

Contiene 8x PAM8302A (mono Class-D filterless)
Cada PAM8302A amplifica 1 canal diferencial

Circuito por PAM8302A:

```
  PCM1690 VOUT+ --[0.1uF]-- IN+
                             PAM8302A     VO+ --- Speaker +
  PCM1690 VOUT- --[0.1uF]-- IN-   (MSOP-8)     VO- --- Speaker -

  +5V ---- VDD ---- SD (enable)
  GND ---- GND
  [10uF + 1uF] bypass en VDD
```

Componentes por IC:
- 2x Cap 0.1uF (coupling entrada IN+/IN-)
- 1x Cap 10uF (bypass VDD baja freq)
- 1x Cap 1uF (bypass VDD alta freq)

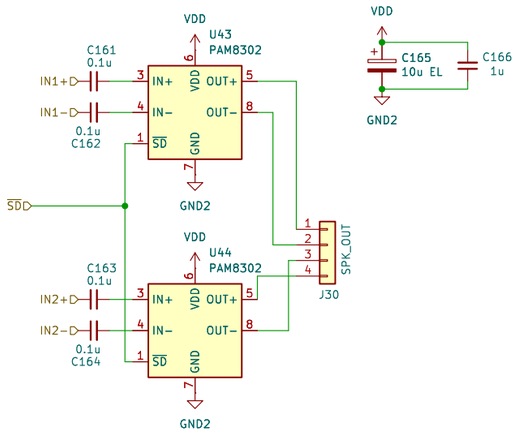
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 Pin 7: GND
 Pin 8: VO-  (salida BTL -)
```

Potencia: ~1.5W @ 8ohm/5V
Entrada: Diferencial verdadera
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LCSC: C113367 (PAM8302AASCR, Diodes Inc)
Precio: ~$0.10/u



Instanciado 4 veces: Bank A/B/C/D
4x PAM8403 = 8 canales

Sheet: /HAT_Amplifiers/ALL_AMPS/Amp_Bank_A15/
File: Amp_Bank.kicad_sch

Title: DS_Amp – Amplifier Bank (8 channels)

Size: A3	Date: 2026-02-05	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 29/29